

YUANKAI TAO

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EDUCATION

Georgia Institute of Technology <i>M.S. in Operations Research, Stewart School of Industrial and Systems Engineering</i> Selected Coursework: Deterministic Optimization, Stochastic Process, Simulation	<i>Aug 2026 — Dec 2027</i>
University of California, San Diego <i>M.S. in Quantitative Finance, Rady School of Management</i> Selected Coursework: Deep Learning, Computational Finance Methods, Data Science for Finance	<i>Sep 2024 — Dec 2025</i> GPA: 3.84/4.0
Zhongnan University of Economics and Law <i>B.S. in Management, School of Accounting and Finance</i> Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming	<i>Sep 2020 — Jun 2024</i> GPA: 3.63/4.0

TECHNICAL SKILLS

Programming & Tools	Python, SQL, MATLAB, MS Office, VBA, L ^A T _E X, Git, Linux, GCP
Quantitative Methods	Statistic & Probability Theory, Time Series Analysis, A/B testing, Advanced Risk Management, Derivatives Pricing, Optimization, Monte Carlo Simulation, Stochastic Process, ML & DL
Skills & Techniques	Data Cleaning, Data Visualization, Feature Engineering, Model Selection, Backtesting, Hyperparameter Tuning, Ensemble Learning, Web Scraping
Libraries & Frameworks	Pandas, Polars, Scikit-learn, PyTorch, Numpy, Scipy, CVXPY, Backtrader, Matplotlib

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. Quantitative Research, Intern	<i>May 2026 — Jul 2026</i> <i>Quantitative Strategy Department Guangdong, China</i>
- Engineered end-to-end commodity futures ML pipeline with 16 features (trend, momentum, volatility, liquidity, volume, open-interest); trained and ensemble LightGBM, XGBoost, CatBoost, and neural networks across 1- to 20-day horizons using rolling out-of-sample validation with both statistical and trading metrics. - Developed a monthly rebalanced cross-sectional futures strategy trading top-q commodities with using LightGBM return probabilities with 120-day rolling retraining window; implemented in Backtrader achieving 1.89 Sharpe ratio after transaction costs, 16.79% annualized return, and 25.51% excess return. - Built a dynamic VaR/CVaR risk measurement system for metal futures; constructed 17 risk models combining 5 volatility models (GARCH, EWMAV) with 3 distribution assumptions (Normal, Student-t, Skewed-t, Filtered Historical Simulation); introduced semi-variance slicing method to isolate long/short momentum risks.	
VariFlight Technology Data Analyst, Intern	<i>Jul 2025 — Sep 2025</i> <i>Data Development Department Anhui, China</i>
- Utilized a Python-based stack for web scraping to gather public reports and web pages/images from the Civil Aviation Administration of China (CAAC). - Employed both traditional and deep learning-based OCR/image recognition tools including pytesseract, Camelot, etc. to extract unstructured data from 600+ non-formatted pdf reports and 500+ webpages and automated the transfer from raw data into formatted data files as input for further local LLM training. - Constructed dashboard on several crucial indexes(e.g., transportation turnover, passenger turnover), and visualized the data for better insights.	
Haitong Securities Investment Analyst, Intern	<i>Jan 2024 — May 2024</i> <i>Investment Banking Department Shanghai, China</i>
Conducted due diligence for IPO projects using Wind financial terminal, verifying client creditworthiness and the reasonableness of fluctuations in sales and procurement figures.	

- Utilized Wind terminal to conduct a review of key expenditure items and their proportions across peer companies, including cost of research and development, and provide preliminary recommendations regarding key points of focus for the declaration during IPO.

Ernst & Young

Audit, Intern

Dec 2022 — Feb 2023

Financial Service Office (FSO) | Beijing, China

- Assisted the client, Hyundai Motor Finance Company, in auditing financial data related to inter-company and investment transactions. Conducted variance analysis: investigated significant deviations, and reviewed supporting documents for further information. Drafted reports on anomalies with recommendations.
- Utilized an automated pipeline to clean financial tables from 220+ csv files and generated standardized audit reports with the processed outputs.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill

Mar 2025 — Jan 2026

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

- Engineered a large-scale data pipeline (Python, PostgreSQL, PostGIS) to conduct spatial computations on a 76-million-observation spatial dataset(40+ variables) on Google Cloud Platform (GCP), using BigQuery and Google Compute Engine (GCE).
- Implemented parallelized spatial computation workflows to calculate comparable housing variables across multiple distance ranges with property-specific filtering criteria, achieving approximately 50× computational speedup.
- Built and managed a SQL database (964.5M records, over 100GB) with Python and DuckDB on UNC's Longleaf high-performance computing cluster; executed complex geospatial queries to extract population statistics with time-window and property-related constraints.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition

Silver Medal

- Processed 47M minute-level observations spanning 1,669 trading days; engineered cross-sectional, lagged, rolling, and market-regime features from 79 anonymized variables and 9 responder targets.
- Trained and ensembled LightGBM, CatBoost, XGBoost, MLP, and TabNet models using time-aware validation; achieved 94th place (Top 2.5%) of 3,757 teams on private leaderboard.

UCSD / FMSbonds, Inc. 2025 Capstone Project

Aug 2025 — Dec 2025

Team Lead | *High-Performance Bond Pricing Engine Optimization*

- Completed the core functions related to risk detection for the library.
- Designed a caching system to memorize high-frequency cash flow queries, eliminating the overhead on redundant generation of cash flows object, reducing the per bond solving time from 1.17 ms to 0.13 ms, achieving a 9× performance increase.
- Rewrote the core yield-to-worst numerical solver using Numba JIT, reducing the computation time from 1.57 ms per bond to 7.5 μs per bond, achieving a 210× speedup.

UCSD CSE 251B Class Competition 2025

CSE251B Final Project, UCSD

Trajectory Prediction Challenge (based on Argoverse 2 Motion Forecasting Dataset)

- Performed data preprocessing, especially data augmentation on raw data, including 50% probability coordinate rotation, flipping, and Gaussian noise addition.
- Architected a OneStepLSTM model with rolling features for autoregressive prediction, leading to a performance improvement comparing to traditional Decision-Tree models(test metric improved from 10.22 to 8.50).

Text Analysis and BERT Fine-tuning on SEC 10-K reports

MGTF 423 Final Project, UCSD

Team Lead | *In-depth Text Analysis & Human-in-the-Loop BERT Fine-Tuning on 20.5M Text Rows*

- Conducted in-depth exploratory analysis on *Fluence Energy, Inc.*, using NER, frequency profiling, word clouds.
- Implemented a Human-in-the-loop iterative fine-tuning to address label scarcity; Trained a BERT model on a seed dataset with 3 rounds of iterations, achieving a 79% accuracy rate in quantifying the uncertainty tone in

disclosures.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate)	Coursera, DeepLearning.AI
Algorithms and Data Structures Certificate (6-Course Certificate)	Coursera, UC San Diego
Beta Gamma Sigma (Top 20% of graduating class)	UCSD, 2024 — 2025
WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%)	2024
Special Scholarship in Innovation and Entrepreneurship	ZUEL, 2022 — 2023